

Mobile Move-Out Inspection and AI-Assisted Property Condition Assessment System

Project Proposal

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1. Executive Summary

This proposal recommends the development of a mobile property inspection application for TexasRenters.com. The application will primarily support technicians performing move-out inspections and assist the operations team in identifying property changes, defects, damages, and possible tenant responsibilities.

The current process requires technicians and staff to manually compare photographs taken before tenant occupancy with photographs or observations recorded during move-out. Reviewing these materials, identifying matching rooms, and manually documenting changes can take several hours. As a result, technicians may only complete approximately one or two inspections per day.

The proposed application will replace much of this manual process with a structured, room-based video inspection.

Each property will have an uploaded floor plan. Artificial intelligence will assist in extracting the rooms and inspection areas from the floor plan. The extracted areas will be converted into room tags such as:

- Living Area
- Kitchen
- Dining Area
- Bedroom 1
- Bedroom 2
- Master Bedroom
- Master Bathroom
- Common Bathroom
- Garage
- Backyard

An authorized user will review, edit, and approve the extracted areas before they are used during an inspection.

During the move-out inspection, the technician will select an approved room tag and record a separate video for that room or area. The technician will state the area name and describe visible defects while recording.

The system will then:

1. Upload and securely store each room-specific video.
2. Automatically transcribe the technician's narration.
3. Associate the transcript and video with the selected room tag.
4. Identify defects and condition changes.
5. Compare move-out findings with the documented move-in or baseline condition.
6. Generate an AI-assisted room summary.
7. Produce a structured draft condition-assessment report.
8. Allow authorized personnel to verify possible tenant-caused damage.
9. Support the final assessment of applicable tenant charges.

Artificial intelligence will assist with organizing and summarizing evidence. Final decisions regarding tenant responsibility, normal wear and tear, and financial charges will remain under human review.

The proposed technology stack includes React Native for the mobile application, Supabase and PostgreSQL for backend services and relational data, Cloudflare Stream for video uploading, storage, processing, and playback, a dedicated speech-to-text provider for transcription, and Anthropic Claude for floor-plan analysis, transcript summarization, and condition comparison.

2. Current Business Process

The existing inspection process generally follows these steps:

1. Photographs are taken before a tenant occupies the property.
2. These photographs become the initial or move-in condition record.
3. When the tenant moves out, a technician inspects the property.
4. The technician captures new photographs or records observations.
5. The technician or office team manually reviews the move-in and move-out evidence.
6. Staff identify which photographs belong to the same room or area.
7. The team manually compares the evidence.
8. Defects and damages are manually documented.

9. The team determines whether each condition is:

- Existing damage
- Normal wear and tear
- Owner responsibility
- Possible tenant-caused damage

10. Approved tenant-responsible findings may be included in the final charge assessment.

This process is time-consuming because the reviewer must repeatedly locate matching rooms, compare evidence, and manually describe changes.

The lack of a consistent room-based inspection structure makes the process slower and increases the possibility of missed areas or inconsistent documentation.

3. Business Problem

The current workflow presents several challenges:

- Move-in and move-out evidence must be compared manually.
- Photographs may not follow a consistent room order.
- Reviewers must manually determine which room appears in each photograph.
- Different camera angles make comparison difficult.
- Defects must be manually described.
- Technicians may forget to inspect certain rooms.
- Long review times limit the number of inspections completed per day.
- Reports may use inconsistent terminology.
- Staff may need to contact technicians for clarification.
- Tenant damage assessments may be delayed.
- Property turnover may be delayed while reports are prepared.

Recording one continuous property video may also introduce avoidable problems:

- Walking between rooms creates unnecessary footage.
- Silent or inactive portions increase video duration.
- Larger video files increase upload and storage usage.
- Reviewers must manually search through the full recording.
- Failed uploads may require resending a large file.
- Identifying exact room boundaries becomes dependent on AI interpretation.

The proposed room-based recording model directly addresses these concerns.

4. Project Objectives

The project aims to:

- Reduce move-out inspection processing time.
 - Increase the number of inspections completed per technician per day.
 - Standardize property inspections according to rooms and floor-plan areas.
 - Use AI to extract possible rooms from uploaded floor plans.
 - Allow authorized users to verify and approve extracted room tags.
 - Require technicians to record one video per room or area.
 - Eliminate unnecessary walking and transition footage.
 - Reduce video size and storage usage.
 - Make room-specific videos easier to upload, retry, review, and manage.
 - Automatically transcribe technician narration.
 - Associate every video and transcript with an approved room tag.
 - Generate room-by-room condition summaries.
 - Compare move-out findings with move-in baseline evidence.
 - Identify possible new defects.
 - Link findings to supporting video evidence.
 - Assist staff in determining possible tenant responsibility.
 - Produce consistent and searchable move-out reports.
 - Preserve an auditable property inspection history.
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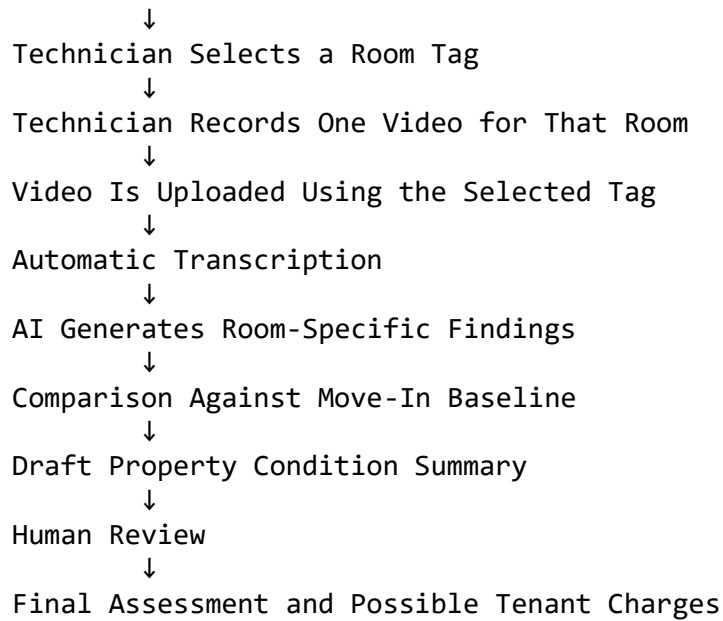
5. Primary Project Feature

The primary feature of the project is:

AI-assisted floor-plan extraction combined with tagged, room-by-room video inspection, automatic transcription, and property condition summarization.

The core workflow is:

```
Property Floor Plan Uploaded
↓
AI Extracts Possible Rooms and Areas
↓
Room Tags Are Generated
↓
Authorized User Reviews and Approves Tags
↓
Technician Opens Assigned Inspection
```



All supporting features should be designed around this workflow.

6. Floor-Plan Extraction and Area Tag Generation

Each property will have an uploaded floor plan.

Supported initial file formats may include:

- PDF
- JPEG
- PNG
- Other common image formats approved during development

After upload, the system will process the floor plan using AI-assisted extraction.

The extraction process may identify:

- Floor labels
- Room names
- Room numbers
- Bedrooms
- Bathrooms
- Living areas
- Kitchen areas
- Dining areas
- Garages

- Laundry areas
- Storage areas
- Exterior areas shown in the plan
- Other labeled inspection areas

Example extracted result

Ground Floor

- Entryway
- Living Area
- Kitchen
- Dining Area
- Common Bathroom
- Garage

Second Floor

- Master Bedroom
- Master Bathroom
- Bedroom 1
- Bedroom 2
- Common Bathroom 2

The extracted areas will be generated as editable tags.

Tag review process

The system will not automatically publish AI-extracted tags without review.

An authorized user will be able to:

- Confirm a room name
- Rename a room
- Delete an incorrect room
- Add a missing room
- Merge duplicate rooms
- Assign the room to a floor
- Reorder the inspection sequence
- Mark an area as required
- Mark an area as optional
- Add an exterior inspection area

Example:

AI-extracted label:

“Bed Room”

Approved tag:
“Bedroom 1”

Once approved, these tags become the official inspection areas for the property.

7. Why One Video Per Room Is Recommended

The system should use one video per approved room or inspection area rather than one continuous video for the full property.

This is the recommended recording method for the MVP and production workflow.

7.1 Reduced unnecessary footage

A continuous video includes:

- Walking between rooms
- Opening doors
- Repositioning the camera
- Waiting for access
- Silent sections
- Repeated explanations
- Accidental inactive recording

These portions do not contribute directly to the inspection evidence.

Recording one room at a time eliminates much of this unnecessary footage.

7.2 Reduced storage consumption

Shorter, focused videos reduce:

- Total recorded minutes
- Cloud storage usage
- Video-processing usage
- Upload duration
- Transcription duration
- Streaming delivery usage
- AI-processing input

This creates a cleaner and more cost-controlled architecture.

7.3 Easier manual review

A reviewer can directly open:

Bedroom 1 Video
Kitchen Video
Master Bathroom Video
Garage Video

The reviewer does not need to search through a long property recording to locate a specific defect.

7.4 More reliable room association

The selected room tag will be attached to the video before recording or upload.

The system will therefore know that:

Video ID: abc123
Room tag: Bedroom 1
Property: Property 045
Inspection: Move-Out Inspection 2026-078

AI will not be solely responsible for determining where one room ends and another begins.

7.5 Easier upload recovery

When one room video fails to upload, only that video must be retried.

The rest of the completed room videos remain unaffected.

7.6 Easier reinspection

When evidence for one room is incomplete, the technician only needs to record that area again.

7.7 Better transcription quality

A focused room video provides better context to the transcription and AI systems.

The AI request can include:

Selected room: Master Bedroom
Technician transcript: ...
Known baseline for Master Bedroom: ...

This significantly reduces ambiguity.

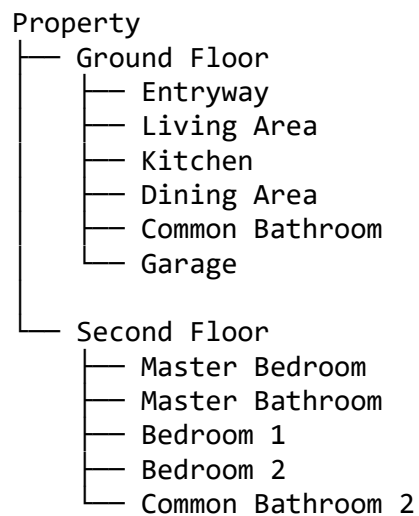
8. Property and Inspection Structure

Each property will contain:

- Property information

- Uploaded floor plan
- Extracted floor-plan data
- Approved floors
- Approved room and area tags
- Required inspection sequence
- Move-in baseline evidence
- Existing documented defects
- Previous inspection history
- Property-specific instructions

Example structure:



Each approved area can have:

- Area name
 - Floor
 - Inspection order
 - Required or optional status
 - Baseline photographs
 - Baseline videos
 - Existing defects
 - Checklist
 - Previous findings
 - Property-specific notes
-

9. Technician Inspection Workflow

9.1 Before the inspection

The technician will:

1. Sign in to the application.
2. View assigned move-out inspections.
3. Open the assigned property.
4. Review property information.
5. Review access instructions.
6. Review the floor plan.
7. Review approved room tags.
8. Download the inspection data for offline use.
9. Verify that all required areas are available.

9.2 During the inspection

For each room or area, the technician will:

1. Select the approved room tag.
2. Review the room's baseline information when appropriate.
3. Start a room-specific recording.
4. Verbally state the room name.
5. Show the overall room condition.
6. Describe visible defects or changes.
7. Capture close-up evidence.
8. Stop the recording before leaving the room.
9. Review the captured video when necessary.
10. Add notes or photographs.
11. Mark the room as completed.
12. Continue to the next room.

Example narration:

"This is Bedroom 1.

There are several deep scratches on the wall beside the closet.

The carpet has a large stain near the window.

The lower closet-door hinge is damaged.

The window and ceiling appear to be in acceptable condition."

9.3 Inspection completion

Before the technician can finalize the inspection, the application will check:

- All required rooms have a video
 - Required checklist items are completed
 - Skipped rooms include a reason
 - Failed recordings are resolved
 - Videos are saved locally
 - Upload statuses are visible
-

10. Room Tag and Video Relationship

Each video will be linked to:

- Organization
- Property
- Inspection
- Floor
- Approved room tag
- Technician
- Recording date and time
- Local file reference
- Cloudflare video identifier
- Upload status
- Processing status
- Transcript
- Findings

Example record:

Property: 1458 Oak Ridge Drive
Inspection: Move-Out Inspection
Floor: Second Floor
Area tag: Bedroom 1
Technician: Technician 004
Recording duration: 2 minutes 48 seconds
Upload status: Completed
Transcription status: Completed
AI status: Ready for Review

This relationship is central to the system architecture.

11. Move-In Baseline and Move-Out Comparison

The system should maintain baseline condition data for each approved property area.

The baseline may contain:

- Move-in photographs
- Move-in videos
- Initial inspection notes
- Existing defects
- Inventory records
- Appliance condition
- Wall condition
- Flooring condition
- Door and window condition
- Fixture condition
- Cleaning condition

Example baseline

Area: Bedroom 1

Wall condition:
Good

Flooring condition:
Good

Closet door:
Functional

Known defect:
Small paint chip beside the light switch

Example move-out observation

Area: Bedroom 1

New observation:
Multiple deep scratches beside the closet

New observation:
Large carpet stain near the window

New observation:

Lower closet-door hinge damaged

Previously documented:

Small paint chip beside the light switch

AI-assisted comparison

Possible new changes:

- Multiple deep wall scratches
- Large carpet stain
- Damage to closet-door hinge

Existing condition:

- Small paint chip beside light switch

Reviewer actions:

- Determine whether the scratches exceed normal wear
- Determine whether the carpet stain is tenant-responsible
- Confirm repair requirement for the closet door

The AI should distinguish between:

- Existing documented defects
 - Possible new defects
 - No material change
 - Normal wear and tear
 - Owner-maintenance issues
 - Missing evidence
 - Insufficient comparison data
-

12. Human Review and Tenant Charge Assessment

AI-generated findings will remain drafts until reviewed.

Authorized reviewers should be able to classify each finding as:

- Existing condition
- No material change
- Normal wear and tear
- Possible tenant-caused damage
- Confirmed tenant-caused damage
- Owner responsibility
- General maintenance issue
- Insufficient evidence

- Requires reinspection
- Rejected AI finding

For possible tenant-responsible findings, reviewers may add:

- Approved defect description
- Responsibility classification
- Repair category
- Estimated cost
- Actual cost
- Supporting baseline evidence
- Supporting room video
- Reviewer comments
- Final charge status

Decision workflow

AI identifies a possible new defect



Reviewer opens the correct room video



Reviewer examines the baseline evidence



Reviewer confirms, edits, or rejects the finding



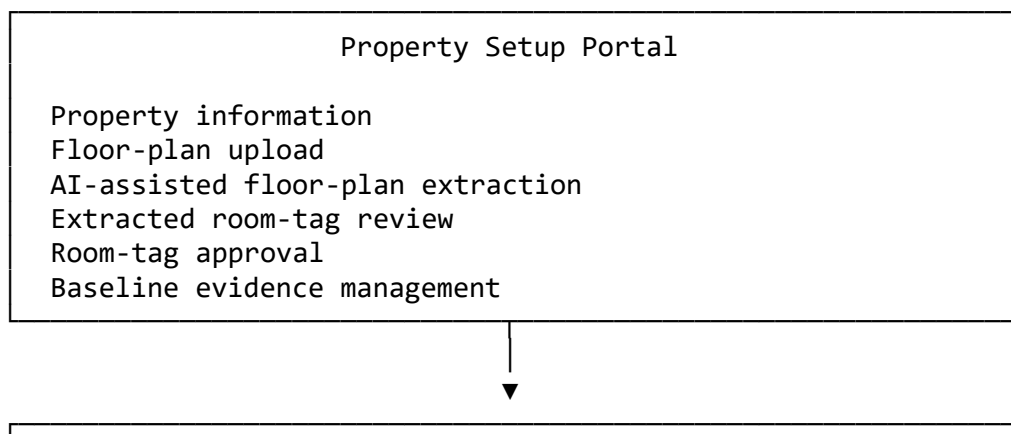
Reviewer classifies responsibility

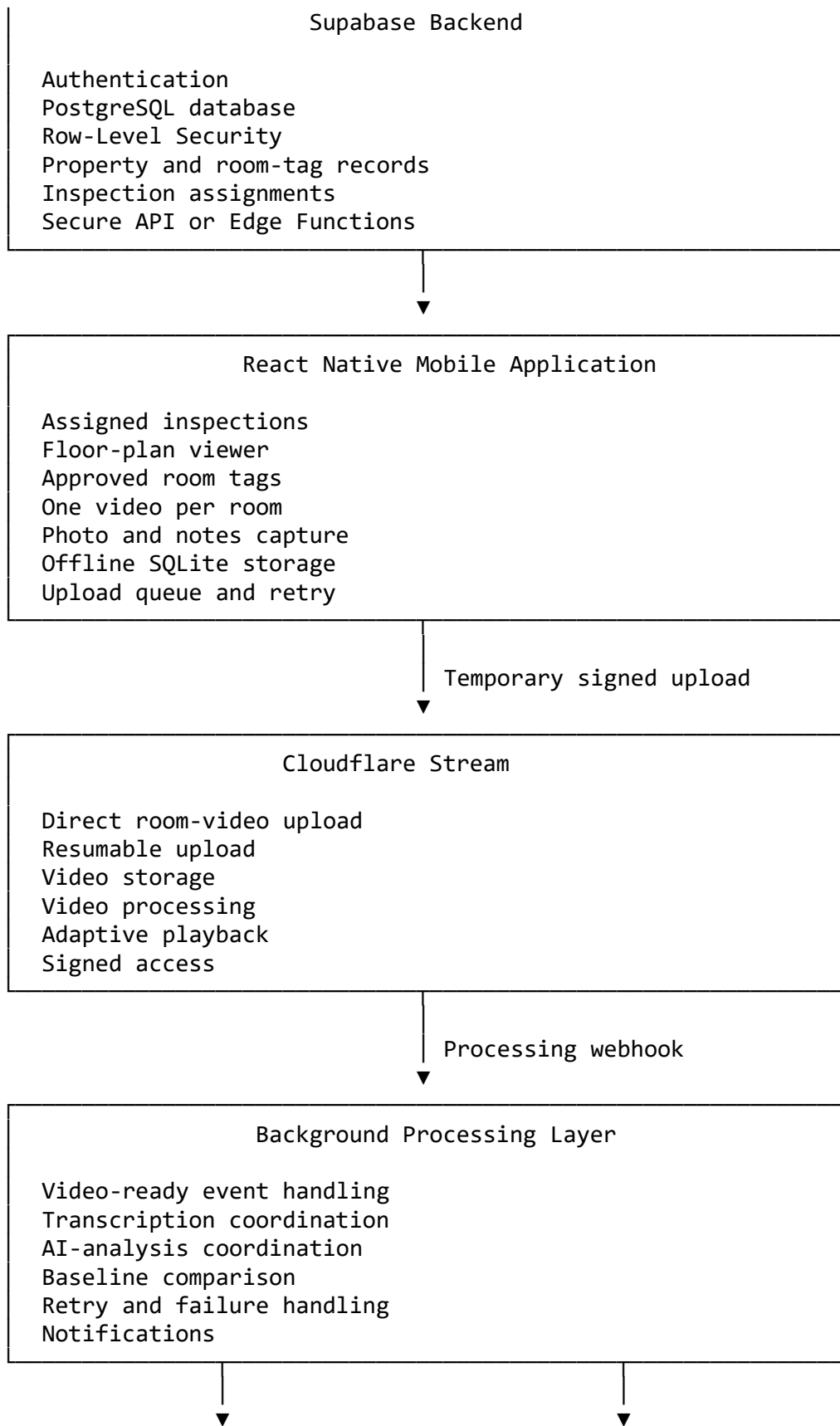


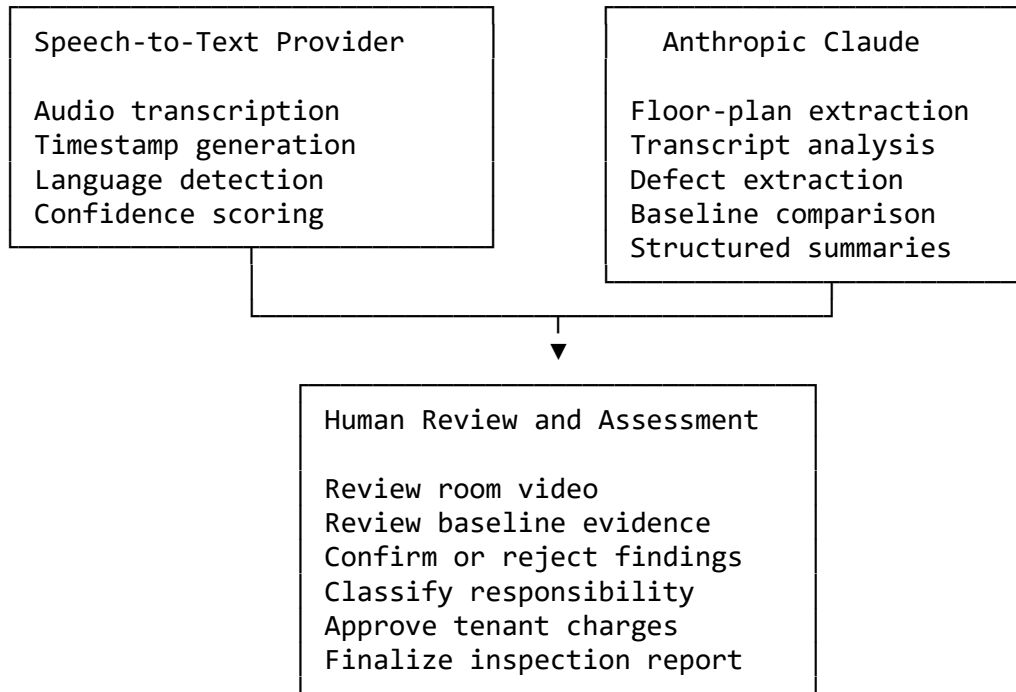
Authorized personnel approve any tenant charge

The AI will not independently approve charges.

13. Proposed System Architecture







14. System Data Flow

Step 1: Floor-plan upload

An authorized user uploads the property floor plan.

Step 2: AI-assisted extraction

The system submits the floor plan for AI-assisted analysis.

The AI returns possible:

- Floors
- Room names
- Room labels
- Inspection areas

Step 3: Room-tag review

An authorized user reviews and corrects the extracted areas.

Step 4: Room-tag approval

Approved areas become available to technicians.

Step 5: Inspection assignment

A move-out inspection is assigned to a technician.

Step 6: Offline synchronization

The mobile application downloads:

- Property information
- Floor plan
- Approved room tags
- Inspection order
- Baseline condition data
- Required checklists

Step 7: Room recording

The technician selects one room tag and records one video for that room.

Step 8: Local storage

The video, room tag, notes, and upload state are stored locally.

Step 9: Direct upload

The application uploads the room video directly to Cloudflare Stream using a temporary signed upload session.

Step 10: Processing notification

Cloudflare notifies the backend when the video is ready.

Step 11: Transcription

The room video is transcribed by the speech-to-text provider.

Step 12: AI analysis

Claude receives:

- Approved room tag
- Transcript
- Timestamped segments
- Baseline condition
- Existing known defects
- Allowed classifications

Step 13: Structured finding generation

Claude returns structured room findings.

Step 14: Human review

The reviewer opens the room-specific video and compares it with the baseline.

Step 15: Final report

Approved findings are included in the move-out condition report and possible charge assessment.

15. Recommended Technology Stack

15.1 React Native

Purpose: Mobile inspection application

Why React Native

- Shared Android and iOS codebase
- Camera and video support
- Local file access
- Offline capability
- Background processing support
- Mature ecosystem
- Lower maintenance cost than separate native applications

15.2 Expo Development Build

Purpose: React Native project tooling and build management

Why Expo

- Faster project setup
- Simplified native configuration
- Camera, file-system, notification, and secure-storage support
- Easier Android and iOS builds
- Support for custom native modules

An Expo development build is recommended because resumable upload and advanced media handling may require native capabilities beyond Expo Go.

15.3 TypeScript

Purpose: Mobile and backend development language

Why TypeScript

- Improves type safety
 - Reduces runtime errors
 - Supports shared mobile and backend models
 - Improves API reliability
 - Makes inspection states easier to manage
 - Works well with React Native, Node.js, and Supabase
-

15.4 Expo Router

Purpose: Application navigation

Why Expo Router

- File-based routing
 - Clear screen organization
 - Authenticated route groups
 - Role-specific navigation
 - Deep-link support
 - Strong Expo integration
-

15.5 SQLite

Purpose: Offline mobile database

Why SQLite

SQLite will store:

- Assigned inspections
- Property information
- Approved room tags
- Floor-plan data
- Inspection progress
- Local video references

- Notes
- Upload queue records
- Failed upload states

This allows technicians to complete inspections without reliable connectivity.

15.6 TanStack Query

Purpose: Backend data synchronization

Why TanStack Query

- Caching
 - Request retries
 - Loading-state management
 - Stale-data refresh
 - Server-state synchronization
 - Reduced networking boilerplate
-

15.7 Zustand

Purpose: Local workflow and interface state

Why Zustand

It can manage:

- Current room
- Active inspection
- Recording state
- Filter selection
- Upload controls
- Temporary interface state

It is lightweight and easy to maintain.

15.8 React Hook Form

Purpose: Checklists and structured forms

Why React Hook Form

It will support:

- Room checklists
 - Technician notes
 - Baseline forms
 - Finding review
 - Responsibility classification
 - Charge review
-

15.9 Zod

Purpose: Runtime data validation

Why Zod

Zod will validate:

- API responses
- Floor-plan extraction output
- Room-tag data
- Inspection submissions
- Webhook data
- Transcript data
- Claude structured output
- Comparison results

AI responses will never be stored without schema validation.

15.10 Supabase

Purpose: Managed backend platform

Supabase will provide:

- PostgreSQL
- Authentication
- Row-Level Security
- Realtime processing updates
- Backend functions

- Database administration

Why Supabase

- Faster MVP implementation
 - Managed relational database
 - Integrated authentication
 - Database-level authorization
 - Standard PostgreSQL compatibility
 - Reduced infrastructure setup
 - Suitable for future reporting and analytics
-

15.11 PostgreSQL

Purpose: Primary database

Why PostgreSQL

The project contains strongly related data:

- Properties
- Floor plans
- Floors
- Room tags
- Baseline evidence
- Inspections
- Videos
- Transcripts
- Findings
- Review decisions
- Tenant charge items

PostgreSQL is better suited than a document-only database for these relationships and comparison queries.

15.12 Supabase Authentication

Purpose: User authentication and session management

Why Supabase Authentication

- Secure sessions

- Password recovery
- Role integration
- PostgreSQL authorization support
- Reduced custom authentication work

Potential roles include:

- System administrator
 - Property administrator
 - Inspection supervisor
 - Inspection technician
 - Condition reviewer
 - Charge approver
 - Read-only property owner
-

15.13 PostgreSQL Row-Level Security

Purpose: Data authorization

Why Row-Level Security

It will enforce rules such as:

- Technicians can only access assigned inspections.
 - Reviewers can only access authorized properties.
 - Charge approvers only access approved findings.
 - Property owners only access their own property records.
 - Unauthorized users cannot view inspection videos.
-

15.14 Cloudflare Stream

Purpose: Room-video uploading, storage, processing, and playback

Why Cloudflare Stream

- Direct mobile uploads
- Resumable uploads
- Video storage
- Automatic processing
- Adaptive streaming
- Secure signed playback

- Centralized media management

Cloudflare Stream removes the need for separate storage, transcoding, and streaming infrastructure during the MVP.

Why no separate Backblaze storage for the MVP

Using Cloudflare Stream and Backblaze for the same videos would create:

- Duplicate storage
- Additional upload logic
- Synchronization work
- Multiple deletion workflows
- More credentials
- More failure points

Additional archival storage should only be introduced if exact original-file retention or independent evidence storage becomes a formal requirement.

15.15 Resumable Uploads

Purpose: Reliable upload of room videos

Why resumable uploads

- Mobile networks may be unstable.
- Application sessions may be interrupted.
- Uploads may pause when devices lock.
- Technicians may switch between Wi-Fi and cellular service.

Room-based videos make resumable uploads even more manageable because each file is smaller and independently recoverable.

15.16 Dedicated Speech-to-Text Provider

Purpose: Technician narration transcription

The provider should support:

- Timestamped transcription
- Language detection
- Accent handling
- Background-noise handling

- Confidence scores
- Prerecorded media
- Multilingual speech

Potential providers include:

- Deepgram
- Azure AI Speech
- Google Cloud Speech-to-Text
- OpenAI transcription services
- Self-hosted Whisper in a future phase

The final provider should be selected after testing real inspection recordings.

15.17 Anthropic Claude

Purpose: AI-assisted floor-plan analysis and condition summarization

Claude may be used for:

- Extracting room labels from floor-plan text
- Normalizing extracted room names
- Detecting duplicate or unclear room labels
- Organizing transcript observations
- Identifying defects
- Comparing findings with the baseline
- Producing room summaries
- Generating structured JSON findings
- Identifying unclear results for review

Claude will not independently:

- Apply tenant charges
 - Determine legal responsibility
 - Approve repair costs
 - Authorize deductions
 - Replace the human reviewer
-

15.18 Node.js Processing Service

Purpose: Background processing and service coordination

The processing service will handle:

- Floor-plan extraction requests
- Upload-session creation
- Cloudflare webhooks
- Transcription requests
- Claude requests
- Baseline comparison
- Retry handling
- Notifications
- Processing statuses

Node.js is recommended because it supports TypeScript, asynchronous APIs, background jobs, and shared validation schemas.

15.19 Background Job Queue

Purpose: Reliable processing

Example processing states:

FLOOR_PLAN_PENDING
FLOOR_PLAN_EXTRACTING
FLOOR_PLAN_REVIEW_REQUIRED
ROOM_TAGS_APPROVED

VIDEO_PENDING_UPLOAD
VIDEO_UPLOADING
VIDEO_PROCESSING
VIDEO_READY

TRANSCRIPTION_PENDING
TRANSCRIPTION_RUNNING
TRANSCRIPTION_COMPLETED

AI_ANALYSIS_PENDING
AI_ANALYSIS_RUNNING
AI_ANALYSIS_COMPLETED

REVIEW_REQUIRED
PROCESSING_FAILED

A queue is necessary because video, transcription, and AI processing may continue after the mobile application is closed.

15.20 Sentry

Purpose: Error tracking and monitoring

Sentry may track:

- Mobile crashes
- Upload failures
- Synchronization failures
- API errors
- Performance problems
- Application-version issues

Sensitive property and tenant data must be excluded from monitoring payloads.

16. Proposed Database Structure

Users and organizations

organizations
users
organization_members
roles
user_devices

Properties and floor plans

properties
property_units
property_floor_plans
floor_plan_extraction_jobs
property_floors
property_areas
property_area_tags
property_access_instructions

Baseline condition

baseline_inspections
baseline_area_conditions
baseline_media
baseline_findings
property_inventory

Move-out inspections

- inspections
- inspection_assignments
- inspection_areas
- inspection_area_status
- inspection_checklists
- inspection_notes

Media and upload

- inspection_media
- media_upload_sessions
- media_processing_events
- media_area_links

Transcription and analysis

- transcription_jobs
- transcript_segments
- ai_analysis_jobs
- ai_analysis_versions
- condition_comparisons

Findings and tenant assessment

- inspection_findings
- finding_evidence
- finding_reviews
- responsibility_assessments
- tenant_charge_items
- charge_approvals

System operations

- notifications
- audit_logs
- integration_events
- failed_jobs

17. Example AI Finding

```
{  
  "propertyAreaId": "bedroom-1-id",  
  "areaName": "Bedroom 1",  
  "findingType": "possible_new_damage",  
  "category": "flooring",  
  "title": "Large carpet stain",  
  "description": "A large dark stain was shown near the bedroom window.",  
  "baselineCondition": "The move-in baseline indicates that the carpet was in  
good condition.",  
}
```

```
"comparisonResult": "possible_new_damage",
"videoTimestampStart": 46,
"videoTimestampEnd": 61,
"severity": "medium",
"possibleResponsibility": "tenant_review_required",
"confidence": 0.91,
"recommendedReview": "Compare the stain against the move-in photograph and
determine whether professional cleaning or carpet replacement is required."
}
```

18. MVP Scope

Property setup

- Create property
- Upload floor plan
- Process floor plan with AI
- Generate possible room tags
- Review and edit extracted tags
- Approve inspection areas
- Set room inspection order
- Mark rooms as required or optional

Authentication and access

- Secure sign-in
- Role-based access
- Session management
- Row-Level Security

Technician inspection

- Assigned inspection list
- Property details
- Floor-plan viewer
- Approved room tags
- One video per room
- Photo capture
- Notes
- Room completion
- Skip room with reason
- Missing-room warning

Offline operation

- Download assigned inspections
- View approved room tags offline
- View floor plan offline
- Record videos offline
- Save progress locally
- Queue uploads
- Resume uploads

Video processing

- Direct Cloudflare upload
- Resumable upload
- Room-tag association
- Upload progress
- Retry controls
- Secure playback
- Video-processing status

Automatic transcription

- Timestamped transcript
- Language detection
- Transcript status
- Transcript review
- Transcription failure handling

AI-assisted analysis

- Room-specific summary
- Defect identification
- Baseline comparison
- Existing-condition detection
- Possible new-damage identification
- Confidence information
- Structured findings

Human review

- Open room video
- View baseline evidence
- Jump to timestamp
- Approve finding

- Edit finding
- Reject finding
- Add manual finding
- Classify responsibility
- Mark normal wear and tear
- Request reinspection

Final report

- Property summary
 - Room-by-room findings
 - Move-in and move-out comparison
 - Supporting photos
 - Supporting room videos
 - Reviewer decisions
 - Possible tenant charge items
 - Final condition report
-

19. Features Outside the Initial MVP

The following features should be considered later:

- Automatic visual comparison between move-in and move-out images
- Computer-vision damage detection
- Automatic room-boundary detection
- 3D floor-plan reconstruction
- Repair cost estimation
- Vendor assignment
- Tenant-facing portal
- Owner-facing portal
- Accounting integration
- Property-management-system integration
- Automated deduction processing
- Digital tenant acknowledgment
- Geolocation verification
- Video redaction
- Automated measurements
- Predictive maintenance

These features should not delay the core room-based inspection workflow.

20. Security and Privacy Requirements

The system should include:

- Private videos by default
- Encrypted network communication
- Short-lived signed playback URLs
- Role-based access control
- Row-Level Security
- Secure mobile credential storage
- Audit logs
- Webhook verification
- API rate limiting
- Account deactivation
- Remote sign-out
- Retention controls
- Secure deletion
- Procedures for sensitive information captured in videos

Permanent Cloudflare, Anthropic, Supabase service-role, or transcription-provider credentials must never be stored in the mobile application.

21. AI Quality and Safety Controls

The system should:

- Treat floor-plan extraction as a draft.
- Require room-tag approval.
- Treat AI findings as drafts.
- Require human review.
- Store room and video relationships.
- Store supporting transcript segments.
- Store video timestamps.
- Display confidence scores.
- Allow corrections and rejection.
- Validate all AI output.
- Prevent duplicate processing.
- Track prompt and model versions.

- Avoid presenting possible damage as confirmed damage.
- Prevent AI from approving tenant charges.
- Record reviewer decisions in audit logs.

22. Data Retention

The company should approve a formal retention policy.

A possible initial structure is:

Data type	Proposed retention
Uploaded floor plan	Retain while the property is managed
Approved room tags	Retain as part of the property profile
Incomplete local inspection	Until uploaded or manually deleted
Failed temporary upload	7–14 days
Temporary extracted audio	Delete after successful transcription
Move-out room videos	Company-approved retention period
Move-in baseline evidence	Retain while required for tenancy and disputes
Raw transcripts	Same period as related inspection evidence
Final reports	Retain as part of property history
Tenant charge assessments	Retain according to financial-record policy
Disputed inspections	Retain until the hold is released
Audit logs	Retain according to security and compliance policy

23. Development Approach

Phase 1: Workflow discovery

- Review current move-in evidence.
- Review existing floor plans.
- Observe an actual move-out inspection.
- Identify standard room naming rules.
- Define normal wear and tear classifications.
- Gather sample technician recordings.
- Confirm reviewer and charge-approval roles.

Phase 2: Floor-plan proof of concept

- Upload a floor-plan image or PDF.
- Extract visible room labels.
- Generate editable area tags.
- Review and approve tags.
- Save approved rooms to PostgreSQL.

Phase 3: Room-video proof of concept

- Display approved room tags in React Native.
- Select a room.
- Record one room video.
- Save it locally.
- Upload it to Cloudflare Stream.
- Preserve the room-video relationship.

Phase 4: AI-processing proof of concept

- Transcribe the room video.
- Generate timestamped transcript segments.
- Send the transcript and room context to Claude.
- Compare the result with baseline data.
- Store structured findings.

Phase 5: Mobile application foundation

- Authentication
- Offline SQLite storage
- Property assignment
- Floor-plan viewer
- Room list
- Room recording
- Upload queue
- Processing status

Phase 6: Review and reporting

- Room-video player
- Baseline viewer
- Finding review
- Responsibility classification
- Tenant charge review
- Final report generation

- Audit history

Phase 7: Pilot

- Test with a limited technician group.
 - Measure average room-video duration.
 - Measure upload success.
 - Measure transcription quality.
 - Measure AI finding accuracy.
 - Compare inspection processing time with the old workflow.
-

24. Testing Strategy

Floor-plan testing

- Clear floor-plan image
- Low-resolution floor plan
- PDF floor plan
- Multiple floors
- Duplicate room labels
- Unlabeled rooms
- Handwritten labels
- Incorrect AI extraction
- Manual room addition

Recording testing

- One video per room
- Multiple videos for one room
- Video retake
- Skipped room
- Required-room validation
- Application restart
- Device storage shortage

Upload testing

- Interrupted upload
- Slow mobile data
- Wi-Fi switching
- Expired upload session
- Duplicate upload

- Retried room video
- Application closure during upload

Transcription testing

- Clear narration
- Different accents
- Multiple languages
- Background noise
- Technician forgets to state room name
- Room tag conflicts with spoken room name
- Long silence
- Multiple defects

AI comparison testing

- Existing defect
- Possible new damage
- Normal wear and tear
- No material change
- Missing baseline
- Incorrect baseline
- Insufficient evidence
- Conflicting technician statement

Security testing

- Technician accessing an unrelated property
- Unauthorized video playback
- Unauthorized room-tag modification
- Invalid webhook
- Disabled account
- Unauthorized charge approval

25. Success Metrics

Recommended measurements include:

- Average inspections completed per technician per day
- Average videos recorded per inspection
- Average room-video duration
- Average total recorded minutes per property

- Reduction in unnecessary video duration
- Average upload time
- Upload success rate
- Resumed-upload success rate
- Floor-plan extraction accuracy
- Percentage of room tags requiring correction
- Transcription accuracy
- AI finding acceptance rate
- AI finding rejection rate
- Average report-preparation time
- Reduction in manual evidence-comparison time
- Time from inspection completion to finalized report
- Technician satisfaction
- Reviewer satisfaction
- Crash-free session rate
- Average processing cost per inspection

The primary expected result is a significant increase in daily inspection capacity while maintaining clear, room-specific evidence.

26. Remaining Team Alignment Items

The following items still require final business confirmation:

1. Who uploads and approves the property floor plan?
 2. Who approves extracted room tags?
 3. Which file formats must be supported at launch?
 4. Are move-in photographs already organized by room?
 5. Must historical move-in evidence be imported?
 6. Who reviews the AI-generated findings?
 7. Who approves tenant-responsible classifications?
 8. Who approves financial charges?
 9. Does the MVP calculate estimated charges?
 10. Which languages must transcription support?
 11. What is the required video-retention period?
 12. Is exact original-file preservation required?
 13. Must the application integrate with an existing property-management or accounting system?
-

27. Recommended MVP Technology Summary

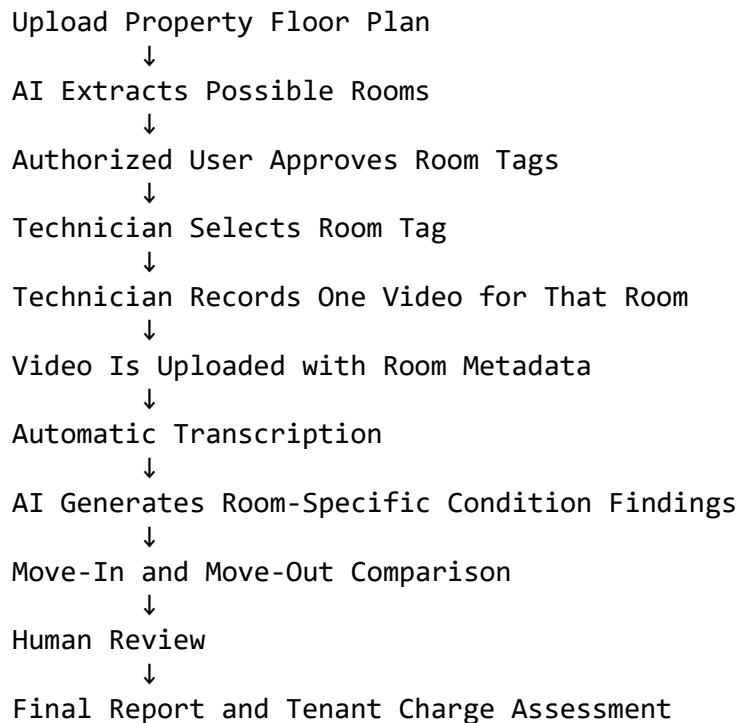
Layer	Recommended technology	Primary reason
Mobile application	React Native	Shared Android and iOS development
Mobile tooling	Expo Development Build	Faster development with native flexibility
Language	TypeScript	Safer and maintainable code
Navigation	Expo Router	Structured application routing
Server data	TanStack Query	Caching, retries, and synchronization
Local state	Zustand	Lightweight workflow management
Offline database	SQLite	Offline inspections and upload tracking
Forms	React Hook Form	Efficient checklists and review forms
Validation	Zod	Safe API and AI data validation
Backend	Supabase	Managed and rapid backend development
Database	PostgreSQL	Relational property and inspection data
Authentication	Supabase Auth	Integrated identity management
Authorization	PostgreSQL RLS	Database-enforced access control
Video platform	Cloudflare Stream	Upload, storage, processing, and playback
Upload method	tus resumable upload	Reliable room-video uploads
Transcription	Provider selected through testing	Dedicated speech recognition
AI processing	Anthropic Claude	Floor-plan extraction and condition summaries
Processing service	Node.js or Supabase functions	Background workflow orchestration
Monitoring	Sentry	Mobile and backend error tracking

28. Final Recommendation

The proposed application should use a room-based inspection architecture.

One continuous property video is not recommended for the primary workflow because it introduces unnecessary transition footage, larger files, increased storage use, more difficult manual review, and more complex AI room identification.

The recommended model is:



This design provides:

- Smaller and cleaner video files
- Reduced storage consumption
- Faster uploads
- Easier failed-upload recovery
- Direct room-video association
- Faster manual review
- More accurate transcript context
- More reliable AI analysis
- Easier reinspection
- Better long-term evidence management

Cloudflare Stream should be used as the primary video platform for the MVP. Supabase PostgreSQL should manage property, floor-plan, room-tag, inspection, transcript, and finding data.

The first implementation milestone should prove the following workflow:

1. Upload a floor plan.
2. Extract possible rooms.
3. Review and approve generated room tags.
4. Display the tags in the mobile application.
5. Select one room.
6. Record one room-specific video.
7. Upload the video.
8. Transcribe the narration.
9. Generate structured findings.
10. Compare the findings with the baseline.
11. Review the findings and supporting room video.

Once this workflow is proven, the project can proceed into full MVP development.